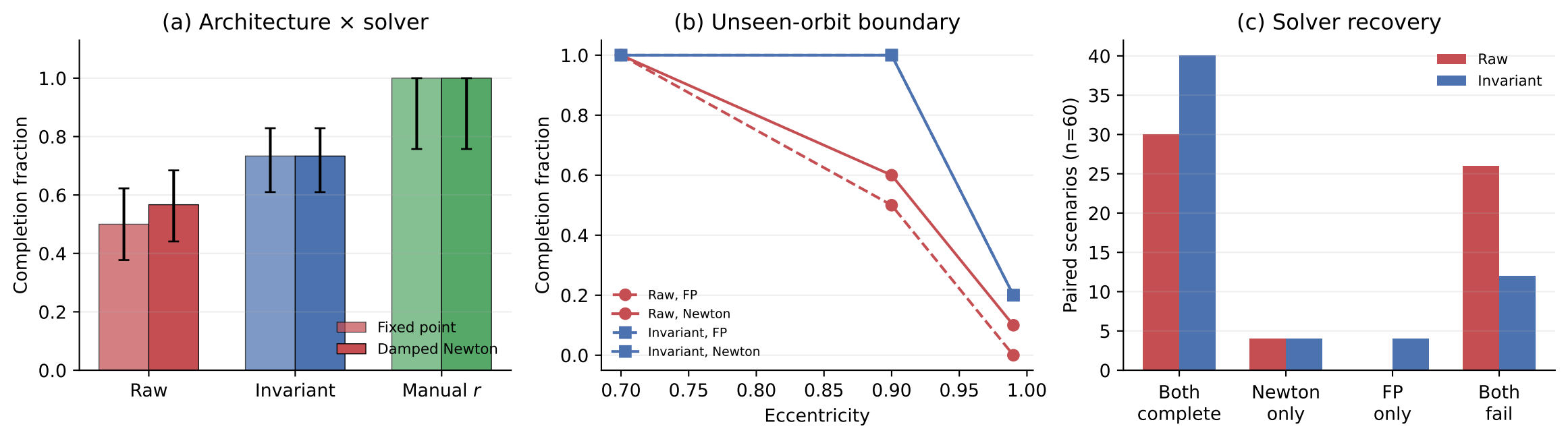
(a) Architecture × solver
Completion fraction
1.0
0.8
0.6
0.4
0.2
0.0
Fixed point
Damped Newton
Raw
Invariant
Manual r
(b) Unseen-orbit boundary
Completion fraction
1.0
0.8
0.6
0.4
0.2
0.0
Raw, FP
Raw, Newton
Invariant, FP
Invariant, Newton
0.70
0.75
0.80
0.85
0.90
0.95
1.00
Eccentricity
(c) Solver recovery
Paired scenarios (n=60)
40
35
30
25
20
15
10
5
0
Raw
Invariant
Both complete
Newton only
FP only
Both fail